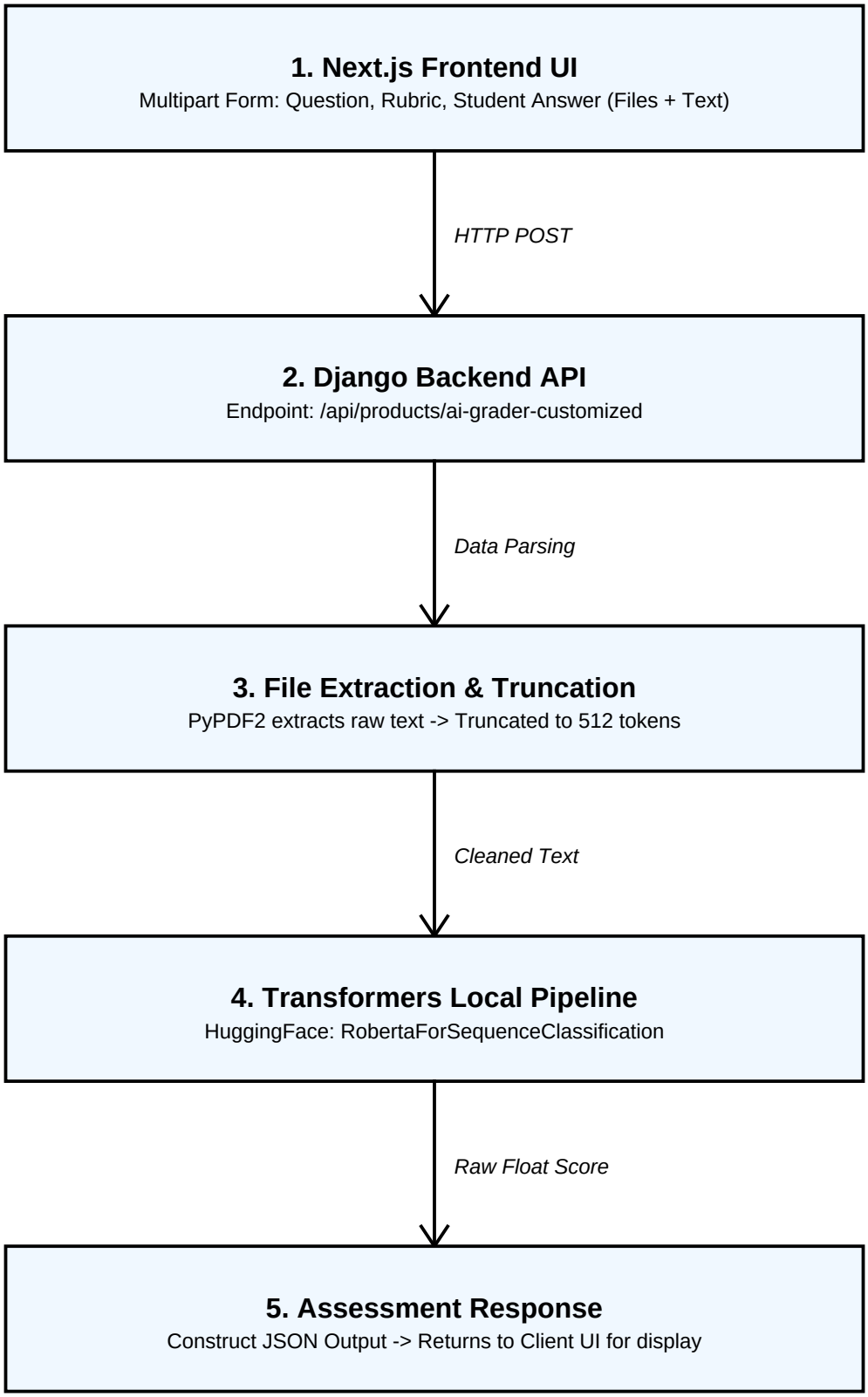
Custom AI Grader - Architecture Flowchart



Architecture Summary:

This flowchart visualizes the secure data pipeline for the Custom AI Grader. The architecture is designed to handle multimodal inputs (text and files simultaneously) from the React UI, extract the content server-side via PyPDF2 in the Django backend, and safely truncate it before passing it entirely within the container boundary to a locally fine-tuned Roberta regression model. By processing the AI grading purely offline with customized transformers models instead of relying on external LLM services like Groq, the application ensures intellectual property privacy and strict grading criteria control.